CS & IT ENGINEERING

DIGITAL LOGIC

Combinational circuit





By- CHANDAN SIR

Lecture No. 02

Recap of Previous Lecture



BOOLEAN ALGEBRA ✓

- D-MORGAN'S LAW
- DUAL & SELF DUAL
- QUESTIONS PRACTICE

✓ K-MAP

Topics to be Covered



comparator

- Comparator (MUX)
- QUESTIONS PRACTICE
- DISCUSSION

Topic : COMPARATOR

DESIGNING OF COMBINATIONAL CIRCUIT

Step 1. Find the number of inputs and outputs.
Step 2. Write the truth table.
Step 3. Write the logical expression.
Step 4. Minimize the logical expression.
Step 5. Hardware implementation.

Topic : COMPARATOR

1. Design a one-bit comparator



Step 1.

Step 2. Truth Table

A	B	(A>B) X	(A<B) Y	(A=B) Z
0	0	0	0	1
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

Topic : COMPARATOR

DESIGNING OF COMBINATIONAL CIRCUIT

Step 3. Logical expression

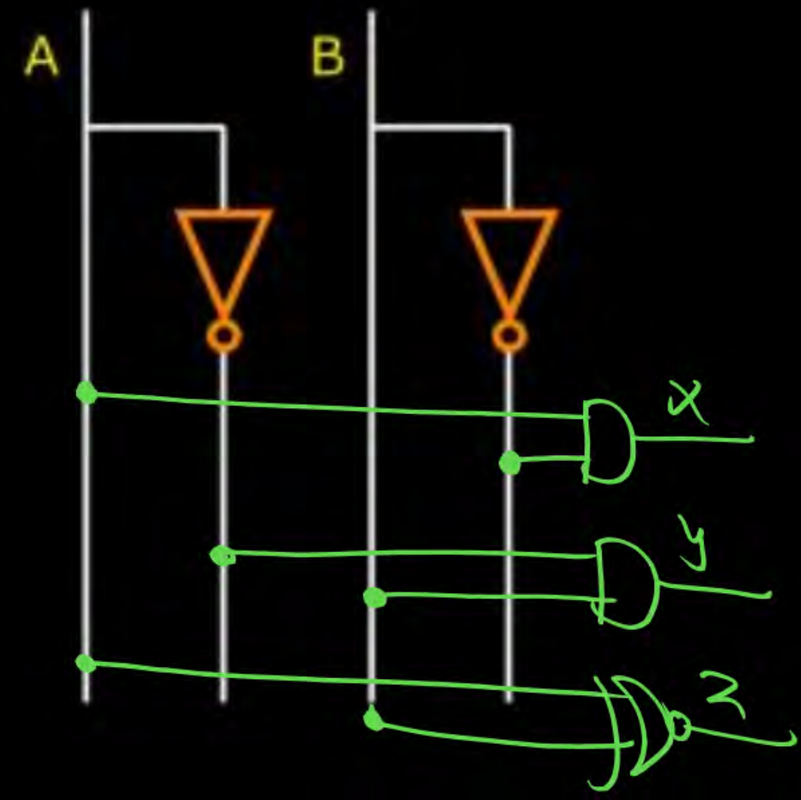
$X(A > B) = A\bar{B}$

$Y(A < B) = \bar{A}B$

$Z(A = B) = A \odot B$

Step 4. Minimization

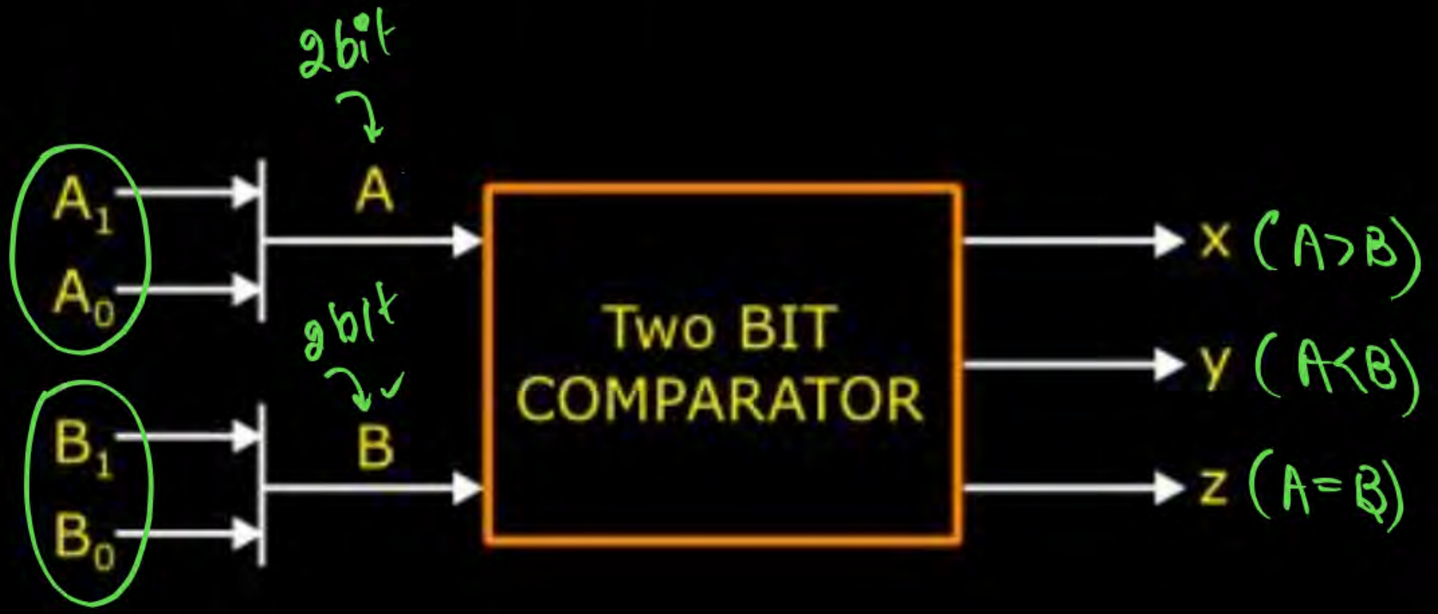
Step 5. Hardware implementation



Topic : COMPARATOR

2. Design a two-Bit comparator?

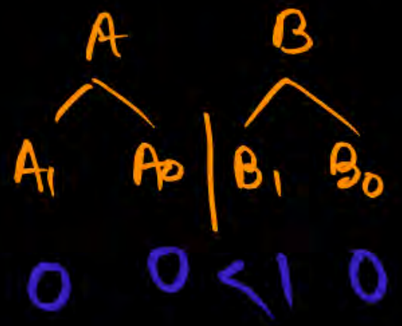
Step 1.



Topic : COMPARATOR

DESIGNING OF COMBINATIONAL CIRCUIT

Step 2. Truth table



	A		B		(A>B)	(A<B)	(A=B)
	A_1	A_0	B_1	B_0	x	y	z
0	0	0	0	0	0	0	1
1	0	0	0	1	0	1	0
2	0	0	1	0	0	1	0
3	0	0	1	1	0	1	0
4	0	1	0	0	1	0	0
5	0	1	0	1	0	0	1
6	0	1	1	0	0	1	0
7	0	1	1	1	0	1	0
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	0	1
11	1	0	1	1	0	1	0
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1		0	1

Topic : COMPARATOR

DESIGNING OF COMBINATIONAL CIRCUIT

Step 3. Logical expression

$X(A > B) = \Sigma m(4,8,9,12,13,14)$

$Y(A < B) = \Sigma m(1,2,3,6,7,11)$

$Z(A = B) = \Sigma m(0,5,10,15)$

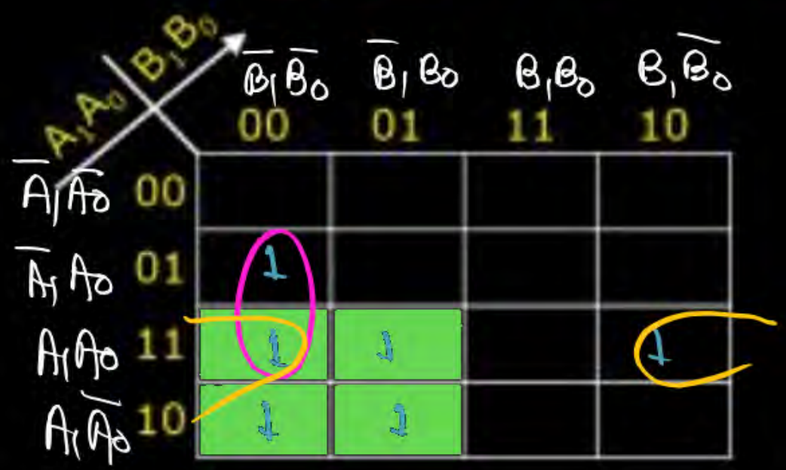
COMPARATOR

DESIGNING OF COMBINATIONAL CIRCUIT

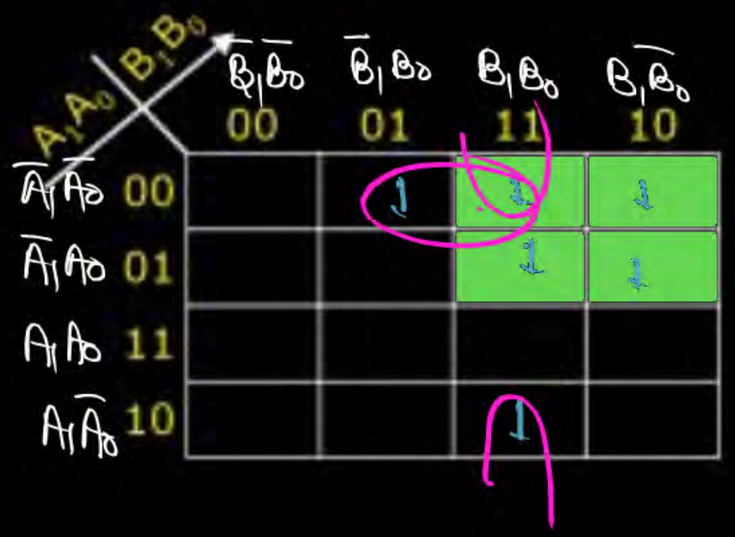
$Y(A<B) = \bar{A}_1 B_1 + (\bar{A}_1 + B_1)\,\bar{A}_0 B_0$

Step 4. Minimization

For X (A>B) $= A_1 \bar{B}_1 + (A_1 + \bar{B}_1) A_0 \bar{B}_0$



For Y

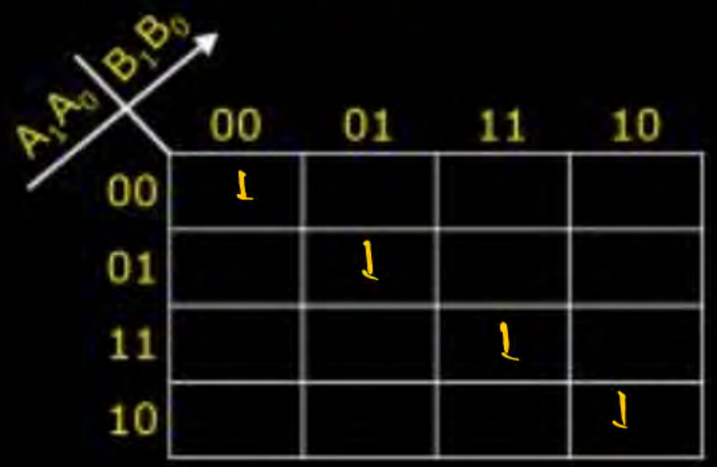


Topic : COMPARATOR

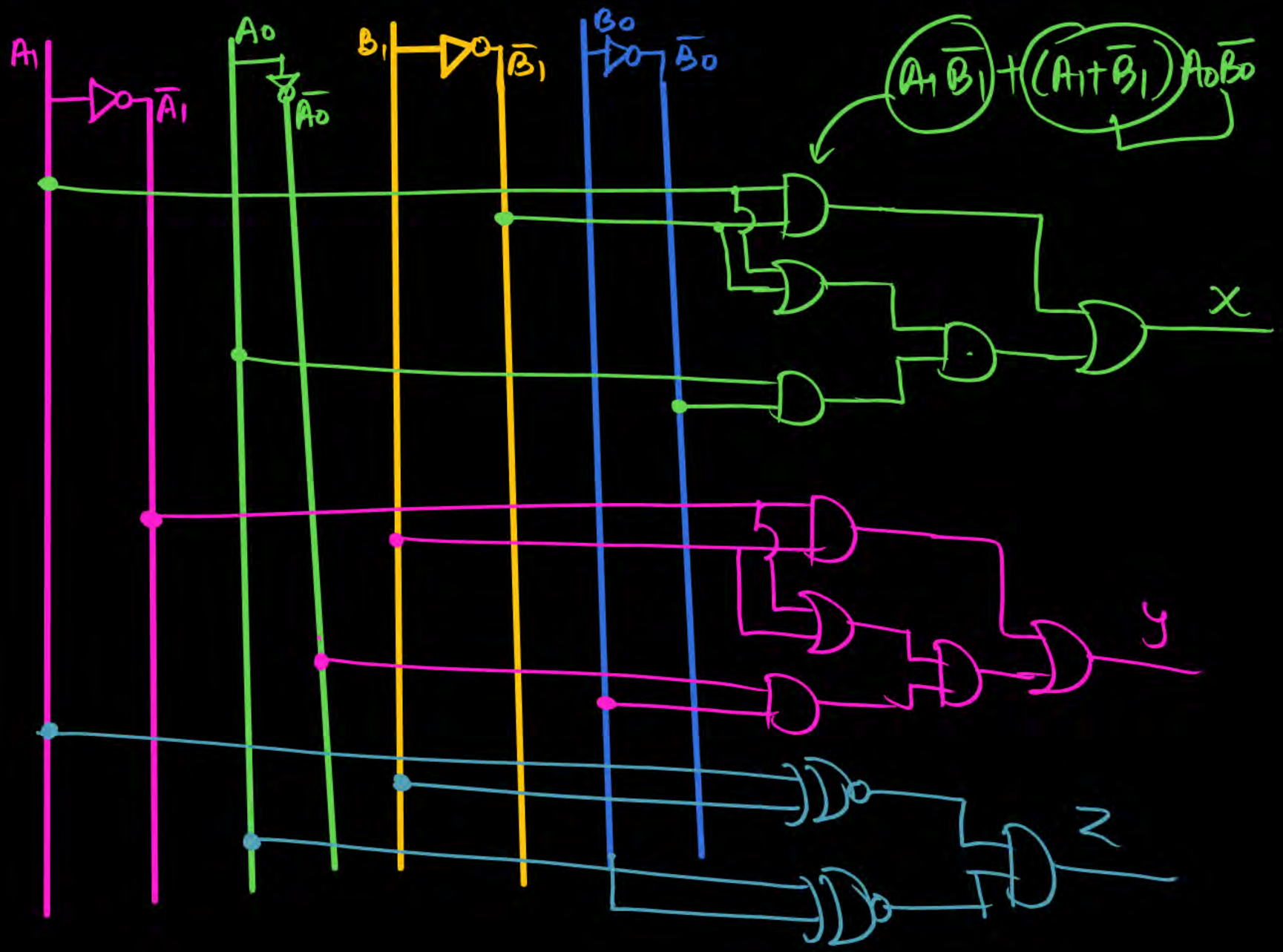
DESIGNING OF COMBINATIONAL CIRCUIT

Step 4. Minimization

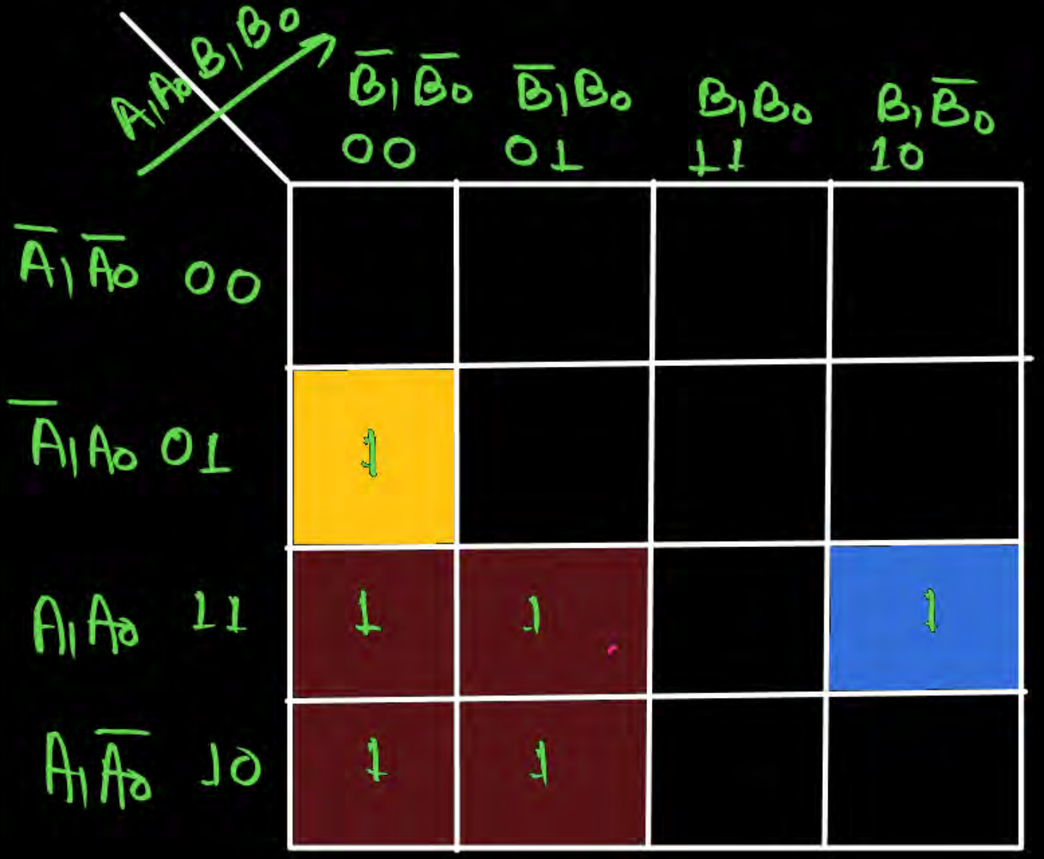
For Z



$$Z(A=B) = (A_1 \odot B_1) \cdot (A_0 \odot B_0)$$


A1
A0
B1
B0
$\bar{A}_1$
$\bar{A}_0$
$\bar{B}_1$
$\bar{B}_0$
$A_1\bar{B}_1 + (A_1 + \bar{B}_1)A_0\bar{B}_0$
x
y
z

X



X $A > B$

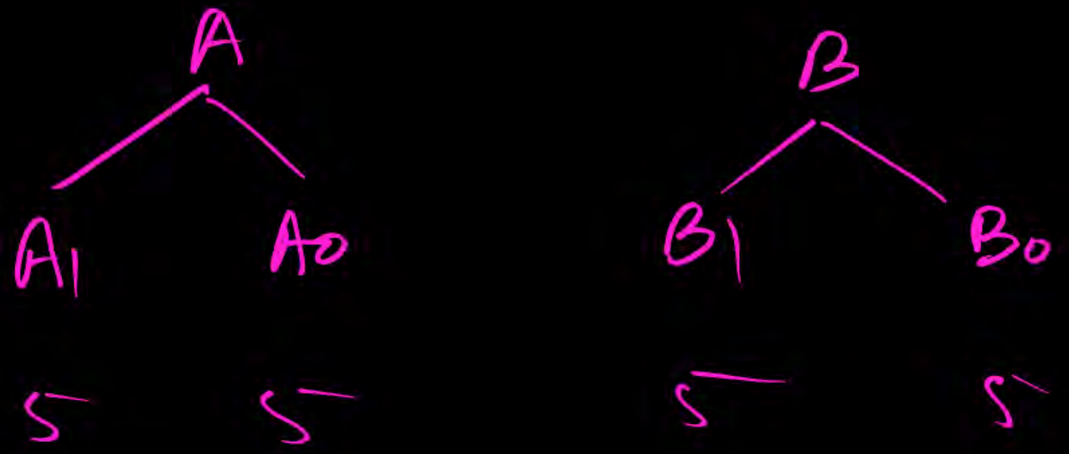
$$A_1\bar{B}_1 + \bar{A}_1A_0\bar{B}_1\bar{B}_0 + A_1A_0B_1\bar{B}_0$$

$$A_1\bar{B}_1 + (\bar{A}_1\bar{B}_1 + A_1B_1)\ A_0\bar{B}_0$$

$$\boxed{A_1\bar{B}_1 + (A_1 \odot B_1)\ A_0\bar{B}_0}$$

$$y(A<B) = \bar{A}_1 B_1 + (A_1 \odot B_1)\, \bar{A}_0 B_0$$

$$z(A=B) = (A_1 \odot B_1)(A_0 \odot B_0)$$



$$X(A>B) = A_1\overline{B_1} + (A_1 \odot B_1)A_0\overline{B_0}$$

$$Y(A<B) = \overline{A_1}B_1 + (A_1 \odot B_1)\overline{A_0}B_0$$

$$Z(A=B) = (A_1 \odot B_1)\cdot(A_0 \odot B_0)$$



$$X(A>B) = A_2\bar{B}_2 + (A_2 \odot B_2)A_1\bar{B}_1 + (A_2 \odot B_2)\cdot(A_1 \odot B_1)A_0\bar{B}_0$$

$$Y(A<B) = \bar{A}_2 B_2 + (A_2 \odot B_2)\bar{A}_1 B_1 + (A_2 \odot B_2)\cdot(A_1 \odot B_1)\bar{A}_0 B_0$$

$$Z(A=B) = (A_2 \odot B_2)\cdot(A_1 \odot B_1)\cdot(A_0 \odot B_0)$$

Topic : COMPARATOR

FOR ONE BIT COMPARATOR

Total condition = 4

Equal condition = 2

Unequal condition = 2

Greater = Less condition = 1

A>B B<A

Topic : COMPARATOR

FOR TWO BIT COMPARATOR

Total condition = 16

Equal condition = 4

Unequal condition = 12

Greater = Less condition = 6

A>B

A<B

Topic : COMPARATOR

FOR THREE BIT COMPARATOR

Total condition = 64

Equal condition = 8

Unequal condition = 56

Greater = Less condition = 28

A>B A<B

A B

0 000 000

63 111 111

Topic : COMPARATOR

'N' BIT COMPARATOR

Total condition = 2^{2n}

Equal condition = 2^n

Unequal condition = $2^{2n} - 2^n$

Greater = Less condition = $(2^{2n} - 2^n)/2$

A>B A<B

Q 4 bit comparator

Total combination = $2^{2n} = 2^{2\times 4} = 2^8 = 256$

Equal combination = $2^n = 2^4 = 16$

Unequal combination = $256 - 16 = 240$

(A>B) or (A<B) = 120

Q

A → 3bit

B → 2bit

$A = A_2 A_1 A_0$

$B = B_1 B_0$

A>B ?

A_2	A_1 A_0	B_1 B_0
0	0 0	0 0
0	0 0	0 1
⋮	⋮	⋮
0	1 1	1 1
1	0 0	0 0
1	0 0	0 1
⋮	⋮	⋮
1	1 1	1 1

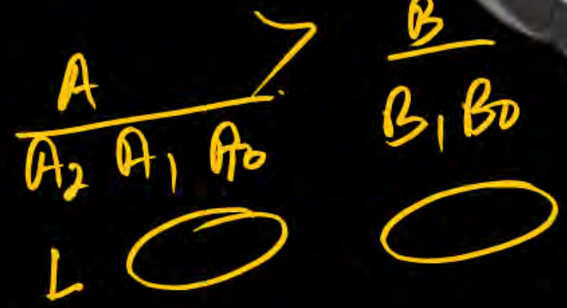
16

16 (A>B)

A<B = 6 Amp

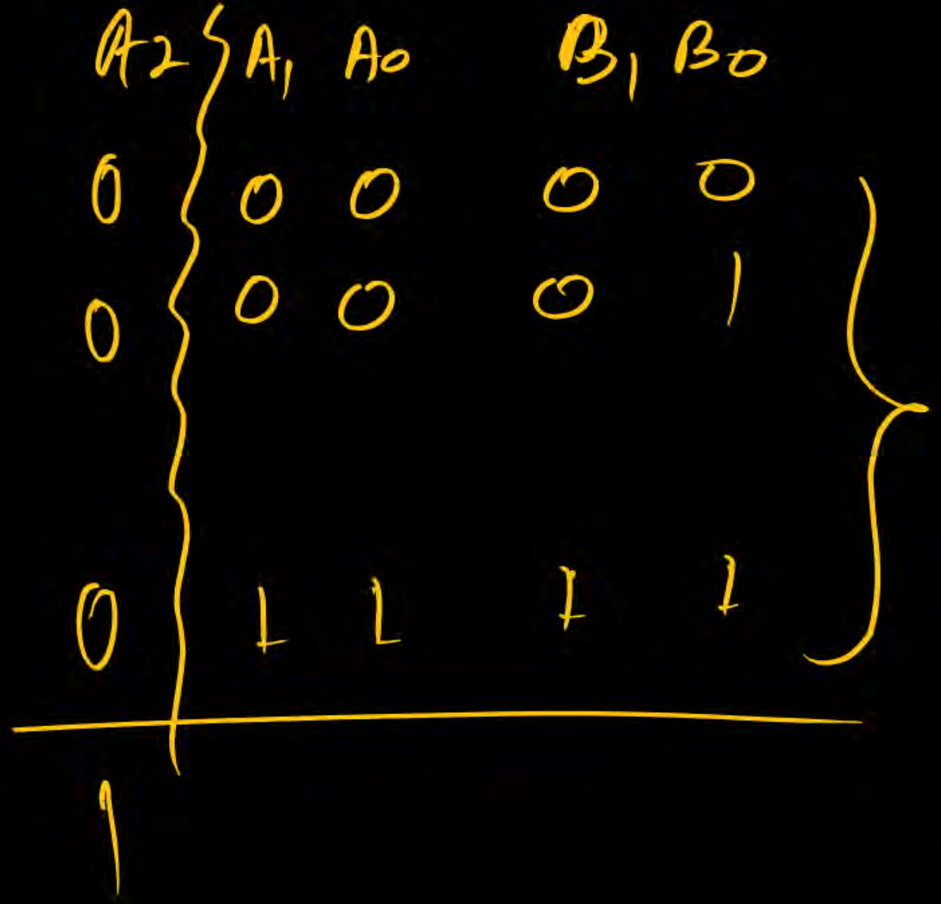
A>B = 22 Amp

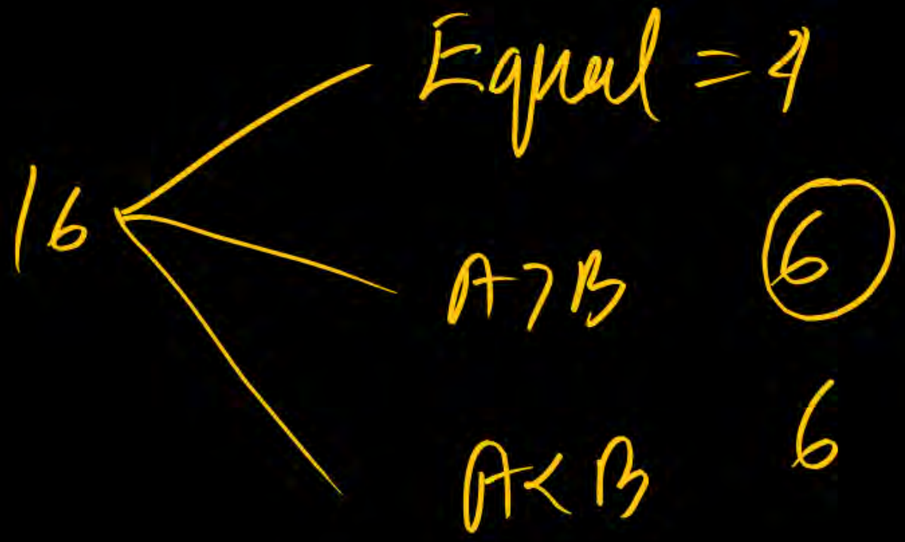
A=B = 4 Amp



A>B ⑥ $\Rightarrow \frac{2^{2n} - 2^n}{2}$

A>B ⇒ 16+6 = 22


A2 A1 A0 B1 B0
0 0 0 0 0
0 0 0 0 1
0 1 1 1 1
1


16
Equal = 1
A > B
6
A < B
6


16
A > B

HW

$A = 5$ bit

$B = 3$ bit

?

$A > B = ?$

$A < B = ?$



2 mins Summary

Comparator MUX

QUESTIONS PRACTICE

DISCUSSION

Thank You GW Soldiers!

